LEVEL UP ACADEMY

Class 10 | GSEB | Science

Pre-Board Mock 3 (GSEB pattern)

Chapter / Topic: Full syllabus

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. Section A (Q1–24) is compulsory: 1 mark each.
2. Section B: answer any 9 of 12 questions (2 marks each). Section C: answer any 6 of 8 (3 marks each). Section D: answer any 5 of 7 (4 marks each).
3. Write balanced equations and draw neat, labelled diagrams wherever needed. Show formulae and units in numericals.
4. This mock follows the reported GSEB blueprint (A 24 · B 18 · C 18 · D 20). Check the latest official GSEB blueprint for changes.

SECTION A (Q1–8): CHOOSE THE CORRECT OPTION — 8 marks

Q1. Which reaction is used in black-and-white photography? **[1]**

(a) Combination of silver and bromine
(b) Displacement of silver by copper
(c) Neutralisation of an acid
(d) Decomposition of silver bromide by light

Q2. Our body works within a pH range of about: **[1]**

(a) 1.0 to 2.0 (b) 4.0 to 5.0
(c) 7.0 to 7.8 (d) 10.0 to 11.0

Q3. Which of these metals is the poorest conductor of heat? **[1]**

(a) Silver (b) Lead
(c) Copper (d) Aluminium

Q4. Which of the following is a ketone? **[1]**

(a) CH_3COCH_3 (b) CH_3CHO
(c) CH_3COOH (d) C_2H_5OH

Q5. Which part of the nephron reabsorbs glucose, amino acids, salts and water? **[1]**

(a) Glomerulus (b) Bowman's capsule
(c) Tubule (d) Ureter

Q6. Which of these is NOT an agent of cross-pollination? **[1]**

(a) Wind (b) Water
(c) Insects (d) Soil

Q7. Which colour of visible light has the longest wavelength? **[1]**

(a) Violet (b) Red
(c) Blue (d) Green

Q8. The SI unit of resistivity is: **[1]**

(a) ohm metre (b) ohm
(c) ohm per metre (d) volt per ampere

SECTION A (Q9–14): FILL IN THE BLANKS — 6 marks

Q9. _____ is used as the reducing agent to obtain zinc from zinc oxide. **[1]**

Q10. Vinegar is a dilute solution of _____ acid. **[1]**

Q11. The hormone insulin is secreted by the _____. **[1]**

Q12. The image formed by a convex mirror is always virtual, erect and _____. **[1]**

Q13. The female reproductive part of a flower is the _____. **[1]**

Q14. A 1 kW appliance running for 1 hour uses 1 _____ of energy. **[1]**

SECTION A (Q15–19): STATE TRUE OR FALSE

5 marks

Q15. All acids contain hydrogen. **[1]**

Q16. Ionic compounds conduct electricity in the solid state. **[1]**

Q17. The human heart has three chambers. **[1]**

Q18. A person with myopia can see nearby objects clearly. **[1]**

Q19. Biodegradable substances can be broken down by decomposers. **[1]**

SECTION A (Q20–24): ANSWER IN ONE WORD

5 marks

Q20. Name the gas given off when baking soda is heated. **[1]**

Q21. Name the alloy of copper and tin. **[1]**

Q22. Name the hormone that regulates the metabolism of carbohydrates, proteins and fats. **[1]**

Q23. Name the phenomenon that makes the sky look blue. **[1]**

Q24. Name the rule that gives the direction of the force on a current-carrying conductor in a magnetic field. **[1]**

SECTION B

answer any 9 of 12 · 18 marks

Answer any 9 questions. 2 marks each.

Q25. Write balanced equations for: (a) sodium + water → sodium hydroxide + hydrogen, (b) zinc carbonate → zinc oxide + carbon dioxide (on heating). **[2]**

Q26. Why do acids not show acidic behaviour in the absence of water? **[2]**

Q27. Give two reasons why aluminium is used to make cooking utensils. **[2]**

Q28. Name the functional group in (a) C_2H_5OH and (b) CH_3COOH. **[2]**

Q29. Give the function of xylem and of phloem. **[2]**

Q30. Give one function each of the medulla and the cerebellum. **[2]**

Q31. What is the difference between a gamete and a zygote? **[2]**

Q32. Give one example each of a dominant and a recessive trait from Mendel's pea experiments. **[2]**

Q33. Give two uses of convex lenses. **[2]**

Q34. Why does a stick partly dipped in water look bent at the water surface? **[2]**

Q35. A current of 0.5 A flows through a 4 Ω resistor. Find the potential difference across it and the power used. **[2]**

Q36. What are CFCs used for, and why are they harmful? **[2]**

SECTION C

answer any 6 of 8 · 18 marks

Answer any 6 questions. 3 marks each.

Q37. Give reasons: (i) Sodium and potassium are never found free in nature. (ii) Gold is found free in nature. (iii) Sodium is stored under kerosene. **[3]**

Q38. Write the names and molecular formulae of the first three alkanes. Why are they called saturated hydrocarbons? **[3]**

Q39. Describe the four chambers of the human heart and the job of each. **[3]**

Q40. Explain phototropism and geotropism, with an example of how each helps the plant. **[3]**

Q41. An object is placed 10 cm in front of a convex mirror of focal length 15 cm. Find the position, nature and magnification of the image. **[3]**

Q42. Resistors of 2 Ω, 3 Ω and 6 Ω are connected in parallel to a 6 V battery. Find (i) the equivalent resistance, (ii) the total current, (iii) the current through the 6 Ω resistor. **[3]**

Q43. Explain the role of the earth wire, the fuse, and the separate 15 A and 5 A circuits in domestic wiring. **[3]**

Q44. What is an ecosystem? Explain the roles of producers, consumers and decomposers, using a pond as the example. **[3]**

SECTION D

answer any 5 of 7 · 20 marks

Answer any 5 questions. 4 marks each.

Q45. Give four properties of ionic compounds, with a reason for each. **[4]**

Q46. Write equations for the preparation of bleaching powder and baking soda, and give two uses of each. **[4]**

Q47. Explain photosynthesis: the raw materials, the three main events, and the role of stomata. **[4]**

Q48. Explain Mendel's monohybrid cross for seed shape (round RR × wrinkled rr) up to F_2, using a checkerboard. **[4]**

Q49. Describe the main parts of the human eye and their functions. What is the power of accommodation? **[4]**

Q50. (a) Derive the formula for the equivalent resistance of resistors connected in parallel. [2]
(b) Two 10 Ω resistors in parallel are connected in series with a 5 Ω resistor. Find the total resistance. [2] **[4]**

Q51. Describe the magnetic field of a current-carrying solenoid. How is an electromagnet made? Give two uses of electromagnets. **[4]**

~ All the best for your boards! ~